

An Introduction to Applied Bayesian Statistics with Stan

2026-08-17



Overview

Course Goals

- Provide the foundation and orientation so that you can
 - Use Stan (or **brms**) in your work
 - Understand and enjoy StanCon talks!

Outline

- About Stan
- Bayesian models and workflow
- A first multilevel model
- **brms** - more multilevel models
- HMC and multi-level models

About Stan

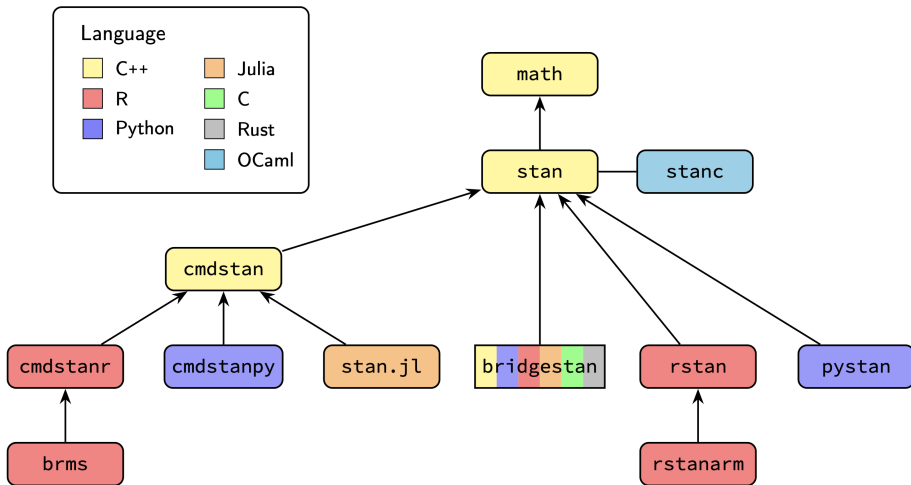
What is Stan?

- A Probabilistic Programming Language
 - A Stan program defines a probability model
- A Collection of Inference Algorithms
 - Exact Bayesian inference using MCMC (NUTS-HMC)
 - Approximate Bayesian inference using ADVI, Pathfinder
 - Point estimates using optimization, Laplace approximation
- Interfaces and Analysis Tools
 - R: [CmdStanR](#), [Posterior](#), [Bayesplot](#), [LOO](#), [RStanARM](#), [brms](#)
 - Python: [CmdStanPy](#), [ArviZ](#)
 - Julia: [Stan.jl](#), [ArviZ.jl](#)
 - Command shell: [CmdStan](#)
 - Programmatic access: [BridgeStan](#)

The Stan Language

- A Stan program
 - Declares data and (constrained) parameter variables
 - Defines log posterior (or penalized likelihood)
 - Computes quantities of interest
- Syntax
 - Influenced by BUGS (plus Java/C++ punctuation)
 - Explicit variable types (like Java/C++, not like Python, R)
 - Named program blocks - distinguish between data and parameters (not like BUGS)
 - Control flow: `if` statements - dynamic branch points (more powerful than BUGS)
- Mathematical operations
 - Stan language includes a very large set of probability distributions and mathematical functions from Stan's math library
 - Efficient, vectorized (almost always)

Stan Ecosystem - Core Software



Stan Ecosystem

- Stan language compiler: `stanc`
 - Translates Stan code to C++
- Programming language interfaces: `CmdStanR`, `CmdStanPy`, `Stan.jl`, (`MatLabStan`)
 - Core methods for model compilation, doing inference, accessing results
- Tools for visualization and validation: `Bayesplot`, `LOO`, `ArviZ`, `Arviz.jl`
 - Evaluate the goodness of fit; make plots to communicate findings
- Higher Level Interface for R: **brms** - Bayesian Regression Models in Stan
 - Model specification via extended R formula, data supplied as R dataframe
 - **brms** translates formula to Stan, creates list of inputs from dataframe, runs NUTS-HMC sampler, checks inferences.
- BridgeStan - algorithm development and exploration
 - Evaluate the log posterior density at a point
 - Constrain/unconstrain model parameters

Stan Documentation

- Stan Documentation Suite
 - [Stan User's Guide](#)
 - [Reference Manual](#) - just the facts
 - [Functions Reference](#) - Stan math library functions available in the Stan language
- Learning Resources
 - [Tutorials](#) - just a few of many
 - [Stan Case Studies](#) - best practices, deep dives for many models
 - [StanCon Talks](#) - past StanCons
 - [Field Guides](#) - education, ecology, epidemiology, cogsci, pharma
- [Stan Forums](#) - your questions answered

Why Stan?

- **Goal:** fit multilevel models for real applications
- *Need:* a way to state a model precisely enough to compute with
- *Solution:* a domain-specific language for probability models

Why Stan?

- **Goal:** fit multilevel models for real applications
- *Need:* a way to state a model precisely enough to compute with
- *Solution:* a domain-specific language for probability models
- *Need:* Efficient MCMC sampler for high-dimensional problems
- *Solution:* Hamiltonian Monte Carlo (flow of Hamiltonian dynamics) (HMC)

Why Stan?

- **Goal:** fit multilevel models for real applications
- *Need:* a way to state a model precisely enough to compute with
- *Solution:* a domain-specific language for probability models
- *Need:* Efficient MCMC sampler for high-dimensional problems
- *Solution:* Hamiltonian Monte Carlo (flow of Hamiltonian dynamics) (HMC)
- *Need:* Support a heterogeneous user base
- *Solution:* Many interfaces (R, Python, Julia), domain-specific examples, case studies, and field guides

Why Stan?

- **Goal:** fit multilevel models for real applications
- *Need:* a way to state a model precisely enough to compute with
- *Solution:* a domain-specific language for probability models
- *Need:* Efficient MCMC sampler for high-dimensional problems
- *Solution:* Hamiltonian Monte Carlo (flow of Hamiltonian dynamics) (HMC)
- *Need:* Support a heterogeneous user base
- *Solution:* Many interfaces (R, Python, Julia), domain-specific examples, case studies, and field guides
- *Need:* Freely available to applied statisticians
- *Solution:* Code and doc open source (BSD, CC-BY)

Why Stan?

- **Goal:** Robust, fully Bayesian inference with HMC
- *Need:* HMC computes with the **joint** log posterior density $\log p(\theta, y)$
- *Solution:* a Stan program **is** a function returning $\log p(\theta, y)$ up to a constant

Why Stan?

- **Goal:** Robust, fully Bayesian inference with HMC
- *Need:* HMC computes with the **joint** log posterior density $\log p(\theta, y)$
- *Solution:* a Stan program **is** a function returning $\log p(\theta, y)$ up to a constant
- *Need:* Gradients of log posterior for HMC
- *Solution:* Reverse-mode algorithmic differentiation

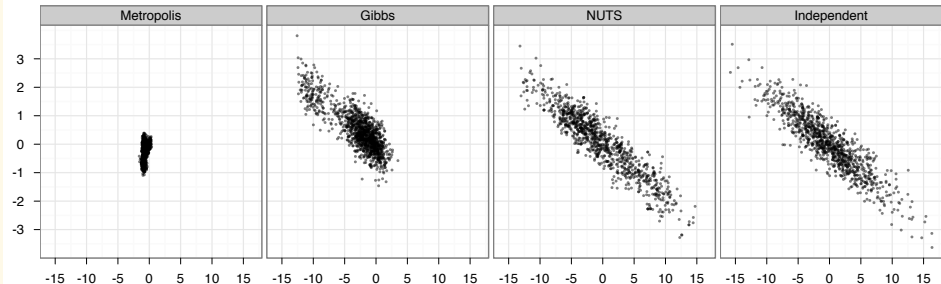
Why Stan?

- **Goal:** Robust, fully Bayesian inference with HMC
- *Need:* HMC computes with the **joint** log posterior density $\log p(\theta, y)$
- *Solution:* a Stan program **is** a function returning $\log p(\theta, y)$ up to a constant
- *Need:* Gradients of log posterior for HMC
- *Solution:* Reverse-mode algorithmic differentiation
- *Need:* HMC computes on the unconstrained scale $\mathbb{R}^N$
- *Solution:* Variable transforms w. Jacobian determinants

Why Stan?

- **Goal:** Robust, fully Bayesian inference with HMC
- *Need:* HMC computes with the **joint** log posterior density $\log p(\theta, y)$
- *Solution:* a Stan program **is** a function returning $\log p(\theta, y)$ up to a constant
- *Need:* Gradients of log posterior for HMC
- *Solution:* Reverse-mode algorithmic differentiation
- *Need:* HMC computes on the unconstrained scale $\mathbb{R}^N$
- *Solution:* Variable transforms w. Jacobian determinants
- *Need:* HMC parameters must be tuned for the posterior geometry
- *Solution:* Tune step size and estimate mass matrix during warmup; determine number of steps on-the-fly (NUTS)

NUTS-HMC vs. Gibbs and Metropolis



- Plot shows 2 dimensions of highly correlated 250-dim normal
- **1,000,000 draws** from Metropolis and Gibbs (thin to 1000)
- **1000 draws** from NUTS (no thinning); 1000 independent draws
- *The No-U-Turn Sampler: Adaptively Setting Path Lengths in Hamiltonian Monte Carlo*, fig. 7

MCMC Sampling Efficiency

- MCMC samples have autocorrelation
- N_{eff} is the number of independent samples with the same estimation power as the N autocorrelated samples.
- When comparing computational efficiency, compare N_{eff} **per second**
- Gibbs and Metropolis have
 - High iterations per second but low effective samples per iteration
 - Both are weak when there is high correlation among the parameters in the posterior
- Hamiltonian Monte Carlo & NUTS
 - Fewer iterations per second than Gibbs or Metropolis
 - But *much* higher N_{eff} per second
 - More robust to highly correlated or differently scaled parameters.

Other Supported Inference Algorithms

Faster, more scalable methods

- Variational Inference

Generate samples from a variational approximation to the posterior distribution

- [ADVI](#) - use stochastic gradient ascent
- [Pathfinder](#) - follow quasi-Newton optimization path

- Optimization: find posterior mode

- MLE - maximum likelihood estimate (of the data given the parameters)
- MAP - maximum a posteriori estimate (of the value of the posterior density on the unconstrained scale)
- Laplace approximation - obtain samples from posterior mode

Bayesian Models and Workflow

Bayesian Data Analysis

- “By Bayesian data analysis, we mean practical methods for making inferences from data using probability models for quantities we observe and about which we wish to learn.”
- “The essential characteristic of Bayesian methods is their **explicit use of probability for quantifying uncertainty** in inferences based on statistical analysis.”

Gelman et al., *Bayesian Data Analysis*, 3rd edition, 2013

Bayesian Workflow

- Set up **full probability model**
 - for **all** observable & unobservable quantities
 - consistent w. problem knowledge & data collection
- **Condition** on observed data (where Stan comes in)
 - to calculate posterior probability of unobserved quantities (e.g., parameters, predictions, missing data)
- **Evaluate** (where Stan sticks around)
 - model fit and implications of posterior
- **Repeat** as necessary (model criticism and comparison)

Ibid.

Priors, Data Distributions, and Likelihoods

- We factor joint models into priors and data generating model.
- The **prior** is what we know about latent parameters before we see the data.
- The **data generating model** generates observable data given parameters.
 - Considered as a function of the parameters for fixed data, the data generating model defines the **likelihood function**.
- Both priors and likelihoods are guided by our **knowledge**.
- Both prior and likelihood are **equally “subjective”**.
 - Conclusions are typically **way more sensitive to the likelihood**.
 - **“Don’t strain at the gnat of prior while swallowing the camel of the likelihood.”** (Andrew Gelman’s blog)

What Determines the Choice of Model?

- **Scientific or domain knowledge** about the process
 - physical, biological, economic, epidemiological mechanisms
- **The way the data were collected**
 - experiments, surveys, observational studies, sampling designs
- **The structure of the data**
 - groups and hierarchies, repeated measurements
 - temporal, spatial, and spatio-temporal dependence
- **The scientific questions we want to answer**
 - the same data may support different models for different questions
- Usually, all of these play a role in determining the starting model; Bayesian workflow guides model criticism and comparison.

Laplace's Model of Birth Rate by Sex

aka, the first Bayesian posterior

Laplace's data on live births in Paris from 1745–1770:

<i>sex</i>	<i>live births</i>
female	241 945
male	251 527

- *Question 1* (Estimation): What is the birth rate of boys vs. girls?
- *Question 2* (Event Probability): Is a boy more likely to be born than a girl?

Laplace computed this analytically; we will use Stan.

Bayesian Inference

A Bayesian model specifies a joint distribution for **parameters and data**:

$$p(y, \theta) = p(y \mid \theta) p(\theta).$$

- $p(\theta)$: **prior** distribution for the parameters
- $p(y \mid \theta)$: **data generating distribution**

After observing y , Bayes's theorem gives the **posterior distribution**:

$$p(\theta \mid y) = \frac{p(y \mid \theta) p(\theta)}{p(y)} \propto p(y \mid \theta) p(\theta).$$

For fixed data y , $p(y \mid \theta)$ is the **likelihood**

posterior $\propto$ **likelihood** $\times$ **prior**

Bernoulli, Binomial Distributions

- The **Bernoulli distribution** has a binary outcome $y \in \{0, 1\}$, with 1 conventionally denoting “success” of a single trial, (e.g., a coin flip).
- The **Binomial distribution** is the number of successes in N trials.
If $y_1, \dots, y_N \sim \text{Bernoulli}(\theta)$, then $(y_1 + \dots + y_N) \sim \text{Binomial}(N, \theta)$, where y is the number of successes in N **independent trials** and where each trial has a $\theta \in (0, 1)$ **chance of success**.
- The **analytic form** is

$$\text{Binomial}(y \mid N, \theta) = \binom{N}{y} \cdot \theta^y \cdot (1 - \theta)^{N-y},$$

where **binomial coefficient** $\binom{N}{y}$ is the **number of sequences** of 0s and 1s of length N with exactly y 1s.

Beta Distribution

- **Beta distributions** are used to **model probabilities** or proportions, thus a Beta-distributed variable has support on the interval $(0, 1)$.
- For parameters $\alpha, \beta > 0$ and a **continuous outcome** $\theta \in (0, 1)$,

$$\text{Beta}(\theta \mid \alpha, \beta) = \frac{1}{B(\alpha, \beta)} \cdot \theta^{\alpha-1} \cdot (1 - \theta)^{\beta-1}$$

(Euler's **Beta function** $B(\alpha, \beta)$ is required to normalize).

- $\alpha - 1$ is prior number of successes, $\beta - 1$ is prior number of failures.
- $\text{Beta}(\theta \mid 1, 1) = \text{Uniform}(\theta \mid (0, 1))$
- **Conjugacy** (Laplace): With a beta prior and binomial data generating distribution, the posterior is a beta distribution.

Laplace's Bayesian Model

- **Data**
 - y : total **number of male live births** (251,527)
 - N : total **number of live births** (493,472)
 - $N - y$: total number of female live births (241,945)
- **Parameter**: $\theta \in (0, 1)$ for the **proportion of male** live births
- **Prior**: $\theta \sim \text{Beta}(1, 1)$
- **Data generating distribution**: $y \sim \text{Binomial}(N, \theta)$
- **Joint density**: $p(y, \theta \mid N) = \text{Binomial}(y \mid N, \theta) \cdot \text{Beta}(\theta \mid 1, 1)$
- **Posterior density**: $p(\theta \mid y, N) = \text{Beta}(\theta \mid y + 1, N - y + 1)$

First Question: Birth Rate of Boys vs. Girls?

- **Posterior** is $\text{Beta}(\theta \mid 1 + 251\,527, 1 + 241\,945)$
- Posterior **mean**:

$$\frac{1 + 251\,527}{1 + 251\,527 + 1 + 241\,945} \approx 0.509708\mathbf{73}$$

- With uniform prior, **maximum likelihood estimate** is the **posterior mode**,

$$\frac{251\,527}{241\,527 + 241\,945} \approx 0.509708\mathbf{82}.$$

- As number of observations approaches ∞ , posterior mean approaches MLE.

Second Question: is a Boy More Likely?

- We want the posterior probability that $\theta > 0.5$:

$$\Pr(\theta > 0.5 \mid y, N) = \int_{0.5}^1 p(\theta \mid y, N) d\theta.$$

- For Laplace's model,

$$\Pr(\theta > 0.5 \mid y, N) \approx 1 - 10^{-42}.$$

- Laplace could compute this analytically because the posterior is a known distribution:

$$\theta \mid y, N \sim \text{Beta}(y + 1, N - y + 1).$$

- For more complicated models, these integrals cannot usually be evaluated analytically.
- **Instead we use Stan**

Laplace's Model in Stan

```
transformed data {  
  int male = 251527;  
  int female = 241945; // total - male  
}  
parameters {  
  real<lower=0, upper=1> theta; // probability male birth  
}  
model {  
  theta ~ beta(1, 1);  
  male ~ binomial(male + female, theta);  
}  
generated quantities {  
  // is a male birth more likely?  
  int<lower=0, upper=1> theta_gt_half = (theta > 0.5);  
}
```


Laplace's Answer (using CmdStanR)

```
births_model = cmdstan_model("laplace_male_births.stan")    # compile model
births_fit = births_model$sample()                          # run Stan's NUTS-HMC sampler
as.data.frame(births_fit$summary())                        # posterior summary (truncated)
```

variable	mean	median	sd	q5	q95
theta	0.5097	0.5097	0.0007213	0.5085	0.5109
theta_gt_half	1.0000	1.0000	0.0000000	1.0000	1.0000

- *Question 1* (Estimation): What is the birth rate of boys vs. girls?
 θ is 90% certain to lie in (0.509, 0.511)
- *Question 2* (Event Probability): Is a boy more likely to be born than a girl?
Laplace “morally certain” boys more prevalent

Monte Carlo Methods

Monte Carlo Methods

- Bayesian inference involves **high-dimensional integrals**.
- **Monte Carlo** methods **estimate** such integrals using probability.
- Specifically, suppose we want to calculate a **posterior expectation**,

$$\mathbb{E}[f(\theta) \mid y] = \int_{\Theta} f(\theta) \cdot p(\theta \mid y) \, dy.$$

with respect to a distribution $\theta \sim p(\theta \mid y)$.

- Suppose we can **randomly sample** $\theta^{(m)} \sim p(\theta \mid y)$
- Then we can use a plug-in **Monte Carlo estimate**,

$$\mathbb{E}[f(\theta) \mid y] \approx \frac{1}{M} \sum_{m=1}^M f(\theta^{(m)}).$$

Stan Samples from the Posterior

- Stan's primary mode of operation is **sampling from a Bayesian posterior**
- Specifically, Stan will draw a **posterior sample** of size M ,

$$\theta^{(1)}, \dots, \theta^{(M)} \sim p(\theta \mid y).$$

- Stan uses **Markov chain Monte Carlo**
 - the sequence of draws is not independent
 - it usually has positive autocorrelation
 - in simple models, autocorrelation can be negative
- Stan's **no-U-turn sampler** (NUTS), uses a form of **Hamiltonian Monte Carlo**
 - NUTS/HMC has the best known scaling in dimension among generic samplers.

Parameter Estimation

- Parameter estimates are posterior expectations (means)

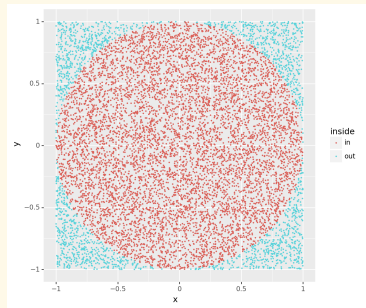
$$\hat{\theta} = \mathbb{E}[\theta | y] = \int \theta \cdot p(\theta | y) d\theta \approx \frac{1}{M} \sum_{m=1}^M \theta^{(m)}$$

- Posterior mean **minimizes expected square error**.
- Posterior median **minimizes expected absolute error**.

Monte Carlo estimation of π

- Estimating $\pi = 3.14 \dots$ via simulation is *the* textbook application of Monte Carlo methods.
- Generate points **uniformly** within the square.
- **Area of circle** is: *proportion within circle times square's area*.
- **Circle's area** is π ; **square's area** is 4.
 - radius $r = 1$ yields area $\pi \cdot r^2 = \pi$
- If $X, Y \sim \text{Uniform}((-1, 1))$, **then**

$$\begin{aligned} 4 \cdot \Pr[X^2 + Y^2 \leq 1] &= \mathbb{E}\left[4 \cdot \mathbf{I}[X^2 + Y^2 \leq 1]\right] \\ &= \pi. \end{aligned}$$



Calculating π with Stan

```
generated quantities {  
  real<lower=-1, upper=1> x = uniform_rng(-1, 1);  
  real<lower=-1, upper=1> y = uniform_rng(-1, 1);  
  int<lower=0, upper=1> in_circle = hypot(x, y) <= 1;  
  int<lower=0, upper=4> pi = 4 * in_circle;  
}
```

- **Simulate** X and Y
- **Indicator** `in_circle` is 1 if (x, y) is in the circle, 0 otherwise.
 - Stan's hypotenuse function: `hypot(a, b) = sqrt(a^2 + b^2)`
- The proportion of points in the circle is $\frac{\pi}{4}$
- **Multiply by 4** to get `pi`, which over multiple draws has **expectation** π .

Fitting the Stan Model with CmdStanPy

```
m = csp.CmdStanModel(stan_file = 'pi.stan')
s = m.sample(iter_sampling=100_000)
s.summary()  # trimmed by hand
```

	Mean	MCSE
pi	3.142900	0.002599

- Our **estimate** is $\hat{\pi} = 3.143$
- 0.0026 is our **standard error** (i.e., standard deviation of estimator error)
- Rough **95% confidence interval** is $\text{mean} \pm 2 \cdot \text{mcse} = (3.138, 3.148)$.
- True value $\pi = 3.141 \dots$ within that interval.

Fitting the Stan Model with Stan Playground

<https://stan-playground.flatironinstitute.org/>

 flatironinstitute / stan-playground ▾



Code Issues 11 Pull requests 1 Agents Actions Projects Security and quality Insights

 main ▾ stan-playground / README.md 

 Go to file T ...

 jsoules Add JOSS paper link to README ✓ bf00b19 · 6 months ago ↻

Preview Code Blame 64 lines (41 loc) · 3.11 KB



Raw  ▾ 

stan-playground

Run Stan models directly in your browser.

[Visit the live site!](#)

JOSS 10.21105/joss.09531



Stan Playground

Accuracy of Markov Chain Monte Carlo

- Monte Carlo Integration computes the exact posterior to within any ϵ
 - i.e., it is *not* an approximation of the posterior like variational Bayes or a Laplace approximation
- Monte Carlo draws are i.i.d. by definition, but MCMC draws are not
- Central limit theorem extends to MCMC: expected error decreases at rate of $\frac{1}{\sqrt{N}}$.
- Need 100× larger sample for each digit of accuracy

Parameter (Co)Variance Estimation

- Posterior **parameter variance** estimates for univariate θ are

$$\begin{aligned}\text{var}[\theta \mid y] &= \mathbb{E}[(\theta - \hat{\theta})^2] \\ &\approx \frac{1}{M} \sum_{m=1}^M (\theta^{(m)} - \hat{\theta})^2\end{aligned}$$

- Posterior **parameter covariance** estimation for multivariate θ is similar,

$$\begin{aligned}\text{cov}[\theta \mid y] &= \mathbb{E}[(\theta - \hat{\theta}) \cdot (\theta - \hat{\theta})^\top] \\ &\approx \frac{1}{M} \sum_{m=1}^M (\theta^{(m)} - \hat{\theta}) \cdot (\theta^{(m)} - \hat{\theta})^\top.\end{aligned}$$

Posterior Event Probabilities

- Posterior **event probabilities** are computed similarly

$$\begin{aligned}\Pr[\text{cond}(\theta) \mid y] &= \int \mathbb{I}[\theta \in A] \cdot p(y \mid \theta) \, d\theta \\ &\approx \frac{1}{M} \sum_{m=1}^M \mathbb{I}[\theta^{(m)} \in A].\end{aligned}$$

- The **indicator function** $\mathbb{I}[\theta \in A]$ evaluates to 1 if θ satisfies the event A , and evaluates to 0 otherwise.
 - i.e., *event probabilities are expectations of indicator functions.*

Posterior Predictive Densities

- Evaluating the **posterior predictive density** is another expectation,

$$\begin{aligned} p(\tilde{y} | y) &= \mathbb{E}[p(\tilde{y}) | y] \\ &\approx \frac{1}{M} \sum_{m=1}^M p(\tilde{y} | \theta^{(m)}) \end{aligned}$$

- **Sampling** from the posterior predictive requires random number generation,

$$\tilde{y}^{(m)} \sim p(\tilde{y} | \theta^{(m)})$$

- **Two forms of uncertainty** compound for posterior predictive,
- **estimation uncertainty** for θ given data y represented by $\theta^{(m)} \sim p(\theta | y)$
- **data generation uncertainty** due to sampling $\tilde{y}^{(m)} \sim p(\tilde{y} | \theta^{(m)})$

Posterior Quantiles

- Posterior **quantiles** (e.g., **intervals** and **median**) just use the empirical quantiles of the draws,

$$\text{quantile}(\theta, \alpha) = \text{sort}(\theta^{(1:M)}) \left[\lceil \alpha \times M \rceil \right]$$

- As α approaches 0 or 1, the number of draws beyond α shrinks and **estimation uncertainty increases dramatically**.
 - e.g., a 2.5% quantile defining the ends of a central 95% interval has around 2.5 times as much uncertainty (coincidental that it's 2.5 and 2.5%) as a mean.

Time for coffee!
